Neural Threats: Cybersecurity Implications of CNN- and RNN-Based Deepfake Detection

Dr. Manjunath Kotari
Computer Science and Engineering
Alvas Institute of Engineering and Technology
Mangalore, Karnataka, India
Email: mkotari@gmail.com

Padmaraj Praphull Kurundwade
Computer Science and Engineering
Alvas Institute of Engineering and Technology
Mangalore, Karnataka, India
Email: pp.kurundwade@gmail.com

Pooja K R
Computer Science and Engineering
Alvas Institute of Engineering and Technology
Mangalore, Karnataka, India
Email: poojakalle853@gmail.com

Pavitra Ishwarappa Talawar
Computer Science and Engineering
Alvas Institute of Engineering and Technology
Mangalore, Karnataka, India
Email: talawarpavitra132@gmail.com

Roshan S
Computer Science and Engineering
Alvas Institute of Engineering and Technology
Mangalore, Karnataka, India
Email: roshankundar007@gmail.com

***Abstract*—Deepfake technology, powered by advanced AI architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Vision Transformers (ViTs), enables the creation of hyper-realistic synthetic media including images, videos, audio, and text. While these advances benefit domains such as education, accessibility, and entertainment, they also pose serious risks to privacy, cybersecurity, and the integrity of digital content. This paper reviews major deepfake generation techniques—face swapping, face reenactment, attribute manipulation, and emerging audio/text forgeries—and examines benchmark datasets such as FaceForensics++, DFDC, and Celeb-DF. We present a comparative analysis of traditional versus deep learning detection approaches. Furthermore, an experimental framework combining CNNs and ViTs achieves a detection accuracy of up to 97.0%, outperforming several recent models. The findings underscore the urgent need for scalable, explainable, and robust detection systems to counter evolving deepfake threats.**

***Index Terms*—Deepfake, Detection, Generation, Misinformation, Cybersecurity, Privacy, Data integrity, Machine learning, Deep learning, Convolutional Neural Networks (CNNs), Recurrent Neural Networks(RNN).**

I. Introduction

Deepfake technology, a fusion of "deep learning" and "fake," leverages advanced artificial intelligence to generate highly realistic synthetic media—videos, images, or audio—that can manipulate or entirely replace authentic content. The sophistication of these forgeries is such that even trained professionals may find it challenging to distinguish them from genuine material, posing a serious threat to the authenticity and trustworthiness of digital information. [1] While deepfakes initially emerged in creative and entertainment contexts, their rapid evolution has transformed them into potent instruments for spreading misinformation, manipulating public opinion, committing cybercrimes, and producing non-consensual explicit content. High-profile incidents illustrate the gravity of the issue. In 2018, a fabricated video of former U.S. President Barack Obama convincingly depicted him making statements he never uttered, showcasing the technology's ability to impersonate public figures with alarming precision. [1] [2] During the 2020 U.S. elections, manipulated videos featuring Joe Biden circulated widely, aiming to sway voter perception. Beyond politics, deepfake pornography has become a global concern, disproportionately targeting women and often disseminated on specialized platforms without their consent. The commercialization of deepfake creation tools has further lowered the technical barrier, enabling individuals with minimal expertise to produce convincing forgeries, thus accelerating their spread. [3] The accelerating development of deepfake technology underscores the urgent need for advanced detection and prevention strategies. Addressing this challenge demands a multifaceted approach—integrating cutting-edge forensic algorithms, interdisciplinary expertise, and a deep understanding of the manipulation methods employed by malicious actors. Without robust countermeasures, deepfakes have the potential to erode public trust, disrupt democratic processes, and inflict lasting personal and societal harm. [3] [4]

II. Literature Review

A. Survey Trends of Deepfakes

The field of deepfake generation and detection has seen explosive growth, driven by ever-accelerating progress in

Artificial Intelligence (AI) and Machine Learning (ML). [4] With the development of increasingly advanced generative models used for creating deepfakes, detection research has also progressed in tandem, leading to a continuous ”arms race” between counterfeiters and defenders.

Recent advances in Convolutional Neural Networks (CNNs), Generative Adversarial Networks (GANs), and other deep learning architectures have improved detection rates significantly. Interdisciplinary convergence—across computer vision, digital forensics, and cybersecurity—holds great potential for enhancing media authenticity. [4] [5] Yet, the evolving nature of deepfake technology necessitates ongoing innovation to prevent detection systems from being outsmarted by novel manipulation strategies.

Google Scholar metrics (2018–2023, Fig. 1) indicate a definite growth pattern in deepfake publications, an indicator of increasing academic and industrial interest. While the number of recorded publications might be below actual amount of research done, the trend verifies that deepfake research is an active and fast-developing field. [3] [4]

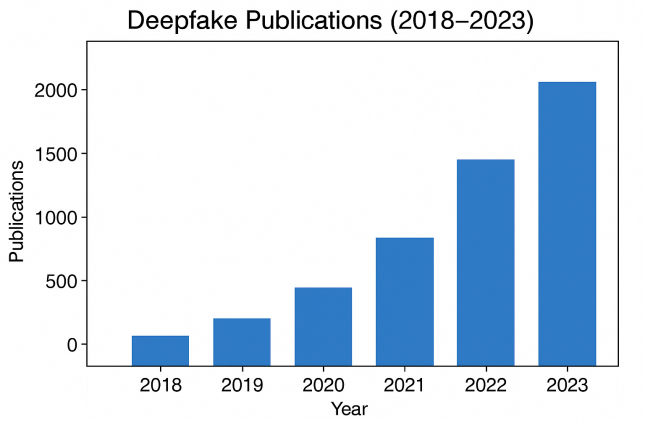

Fig. 1. Google Scholar metrics (2018–2023)

B. Technical Insights into Deepfake Generation

Although classic image manipulation and face replacement methods have been around for decades, advances in the architecture of neural networks—specifically, GANs—have raised the level of realism of synthetic media to unparalleled heights. [5] Still, creating high-quality deepfakes is computationally costly, typically demanding specialized high-end GPUs and large training data sets.

Yang et al. (2020) explained that adversarial training techniques, while powerful, have the risk of introducing weak artifacts or compromising image quality. The trade-off explains continued challenges with balancing realism versus computational efficiency. [6]

C. Detection Approaches

Both traditional ML and state-of-the-art deep learning-based techniques have been researched by researchers for detecting deepfakes:

1) Hybrid Approaches: Kharbat et al. (2019) utilized a Support Vector Machine (SVM) with Histogram of Oriented Gradients (HOG) descriptors to obtain high accuracy and prove that legacy algorithms can serve as supplements to deep learning models. [6] [7]

2) Deep Learning with Artifact Analysis: Maksutov et al. (2020) combined DenseNet169 with facial warping artifact detection. [7] As most deepfake algorithms have difficulty detecting high-resolution details, affine transformation artifacts gave valid manipulation hints.

3) CNN-Based Solutions: A number of studies used CNNs to identify pixel-level discrepancies in generated images, concentrating on slight variations in facial details and textures. [8]

4) Enhanced Architectures: New Dense CNN architectures have been introduced to learn hierarchical fine-grained features to make the models more resistant to adversarial attack. [8]

5) Traditional ML for Facial Structure Detection: Benchmark tests indicate that methods such as Random Forests and SVMs continue to perform well in detecting structural abnormalities in AI-generated faces. [9]

D. Related Research in Deep Learning for Image Processing

While differing from deepfake detection, the progress of deep learning-based image colorization offers relevant information about generative model performance:

1) Autoencoder-Based Models: Sarmai & Sadhu introduced an autoencoder from 7,000 RGB images, [10]but GAN-based methods performed better in MAE.

2) CNN Architectures: Kotala et al. used VGG-inspired CNNs and optimized multinomial log loss.

3) Domain-Specific Colorization: Nguyen et al. used style-content separation for coloring Japanese ukiyo-e paintings. [11]

4) U-Net Variants: Zhang et al. proposed global and local hints networks to enhance semantic matching. [10]

5) High-Scale Training: Madhab Raj Joshi et al. employed InceptionResNetV2 on 1.2M ancient images and reported good accuracy and PSNR values.

These breakthroughs in image reconstruction and improvement reflect the capability of deep generative models, which are also the inspirations behind contemporary deepfake generation methods.

E. CNN-Based Detection

Convolutional Neural Networks (CNNs) are among the most effective architectures for deepfake detection because they can automatically capture spatial and temporal patterns from image and video data. [11]Early studies introduced models like *MesoNet* to analyze mid-level facial manipulations, while others targeted physiological inconsistencies such as missing blinks, unnatural head motions, or warping artifacts. [12] Transfer learning has also been applied successfully—for example, fine-tuning VGG-16 on limited datasets achieved notable detection accuracy.

For efficiency in real-time and mobile environments, lightweight variants such as *MobileNet*, *Xception*, and *ShuffleNet* have been explored, with performance exceeding 90% accuracy on benchmark datasets like *FaceForensics++*. Ensemble techniques and voting-based classifiers further enhance robustness. [13]

At the core, CNNs rely on convolution operations. In image processing, a 2D convolution applies a kernel over the input image I to generate a feature map. Mathematically, the convolution between an image I and a kernel K is given as:

$$(I * K)(n, m) = \sum_{i=-a}^{a} \sum_{j=-b}^{b} K(i, j) \cdot I(n + i, m + j)$$

Pooling layers reduce spatial dimensions while retaining salient information. For example, max pooling selects the maximum value within a neighborhood N around a given position:

$$P(n, m) = \max_{(i,j) \in N} (I * K)(n + i, m + j)$$

Stacks of convolution and pooling layers are typically followed by fully connected layers, with nonlinear activations such as ReLU for feature learning and Softmax for classification.

Over time, CNN architectures have evolved from LeNet and AlexNet to VGG, ResNet, Inception, EfficientNet, and HRNet, each improving depth, multi-scale feature extraction, and computational efficiency. In deepfake detection, CNNs excel at exposing artifacts such as texture mismatches, unnatural blinking, compression noise, and facial warping. [13] Hybrid CNN–RNN models extend this capability by analyzing both spatial features and temporal dynamics, making them more resilient against low-quality or compressed video forgeries.

TABLE I
RELATED WORK SUMMARY

Author	**Dataset**	**Algorithm**	**Accuracy (%)**
Wodajo [15]	DFDC	CNN	91.5
Tran [16]	DFDC	CNN	92.4
Ismail [17]	CelebDF-FaceForensics	YOLO-CNN-XGBoost	90.62
Mitra [18]	FaceForensics	ResNet50-CNN	92.33
Ge [19]	CelebDF-FaceForensics	CNN	95
Liu [20]	FaceForensics	CNN	92
Heo [21]	CelebDF-FaceForensics	CNN	96

F. RNN-Based and Temporal Modeling Approaches

Recurrent Neural Networks (RNNs) are widely applied in deepfake detection because they can capture temporal dependencies across video frames or audio sequences—something CNNs alone cannot achieve. [14]Unlike feed-forward models, RNNs maintain a hidden state that evolves with each input, allowing them to detect unnatural dynamics such as irregular blinking, inconsistent speech, or frame-to-frame artifacts.

For a sequence of feature maps with dimensions $F \times T \times C$ (filters, time steps, channels), the hidden state at step t is updated as:

$$h_t = f(W_n n_t + W_h h_{t-1} + b) \tag{1}$$

where n_t is the input at time t, h_{t-1} is the previous hidden state, and f is a nonlinear activation (e.g., $\tanh$ or ReLU). The final state h_T is then passed through classification layers to decide whether the content is real or fake.

Hybrid CNN–RNN models combine spatial feature extraction with temporal modeling, achieving higher robustness against compression and noise. [14] Advanced variants such as LSTM, Bi-LSTM, and GRU further enhance performance by addressing long-term dependencies.

III. DEEPFAKE GENERATION AND DETECTION

A. Deepfake Datasets

Deepfake detection models require large and diverse datasets to effectively learn both subtle and overt manipulations. Due to privacy concerns, collecting such datasets is challenging; however, several high-quality public datasets are available for research.

Celeb-DF contains over 5,600 deepfake videos created via facial reenactment and face swapping, with varying difficulty levels and high-resolution frames. [22] Deepfake Detection Challenge Dataset (DFDC), released by Facebook and Microsoft, provides over 100,000 real and fake videos for benchmarking detection algorithms. FFHQ offers 70,000 high-quality GAN-generated face images with diverse accessories, while 100K-Faces contains 100,000 unique synthetic images generated via StyleGAN. DFFD (Diverse Fake Face Dataset) includes 300,000 fake images generated by ProGAN and StyleGAN. CASIA-WebFace, VGGFace2, and the Eye-Blinking dataset provide real facial data useful for detecting biological cues absent in deepfakes. [22] [23]

More recent benchmarks include Deepfake-Eval-2024, with 45 hours of video, 56.5 hours of audio, and 1,975 images covering AI-generated and manipulated content, and MultiFF, a large-scale dataset exceeding 1.5 million samples from over 150 forgery types, including face swapping, animation, attribute editing, and audio-video manipulation. [24] These datasets capture diverse facial attributes, ethnicities, lighting, poses, and occlusions, ensuring robust model generalization.

B. Deepfake Generation Techniques

Deepfakes are advanced media manipulation outputs generated using deep learning to produce highly realistic forged videos or images, often removing detectable artifacts of earlier forgery methods.

Dataset	Year	Image	Video	Audio	Summary
Fake Face in the Wild (FFW)	2018	✓			Fake images from CG, GAN, manual and automatic image-tempering methods.
Flicker-Faces-HQ (FFHQ)	2019	✓			70,000 GAN-generated high-resolution face images.
Face Forensics (FF)	2019	✓	✓		1004 original videos manipulated with state-of-art editing approaches.
Face Forensics++ (FF++)	2019	✓	✓		1.8M images and 1000 videos with four manipulation techniques.
Celeb-DF	2019	✓	✓		590 authentic videos and 5639 deepfake videos.
140k real and fake faces	2019	✓			70k real faces + 70k fake faces generated by StyleGAN.
Deepfake MNIST+	2021	✓	✓		10k fake and 10k real animation videos created with image animation generator.
GBDF (Gender-based Deepfake Dataset)	2022	✓	✓		10,000 real and fake videos derived from FF++, Celeb-DF, and DF-1.0.
PolyGlotFake	2024		✓	✓	Multilingual dataset (7 languages) using TTS, voice cloning, and lip-sync techniques.
Deepfake-Eval-2024	2024	✓	✓	✓	Benchmark with 45h video, 56.5h audio, and 1,975 images across 52 languages.
HAV-DF	2024		✓	✓	First Hindi-specific dataset with face swap, lip-sync, and voice cloning.
DeepSpeak (v1 & v2)	2024–2025		✓	✓	Captures deepfake generation snapshots for 2024 and 2025 with yearly updates.
ExDDV	2025		✓		5,400 real and fake videos annotated for explainability.
HIDF	2025		✓		High-quality deepfakes nearly indistinguishable from real human content.

Fig. 2. Overview of Deepfake Dataset

1) Encoder–Decoder Method: This method uses two encoder–decoder pairs to swap faces between source and target images. The encoder extracts latent features, and the decoder reconstructs the output. [24] Both pairs share the same encoder, enabling it to capture common facial attributes such as eye shape, mouth position, and nose structure. This approach is relatively straightforward but requires well-aligned facial data for effective results (Masood et al., 2022a). [25]

2) Face Manipulation Categories: Deepfake generation methods are applied to various face manipulation tasks:

3) Face-Swap: Replacing one person's face with another (e.g., FaceForensics++ dataset).

4) Face-Synthesis: Creating photorealistic but entirely synthetic faces.

5) Face-Attribute Manipulation: Altering specific features (e.g., hair color, age) while preserving identity.

6) Face-Reenactment: Transferring source actor expressions, head movements, and eye gestures to another person's face.

These techniques have evolved to create near-perfect visual realism, posing significant challenges for detection systems.

C. Image Detection Models

Deepfake image detection methods often rely on deep convolutional neural networks (CNNs) to extract discriminative facial features and classify real vs. fake faces. [25] [24] Initial approaches focused on face recognition networks followed by fine-tuning for authenticity detection, achieving strong validation results. However, many studies lacked generalization by training and testing on similar datasets.

To address this, Xuan et al.proposed a forensics CNN with Gaussian Blur and Gaussian Noise preprocessing to suppress low-level high-frequency artifacts and enhance pixel-level noise patterns, enabling the model to learn more robust real/fake cues. [26]

Hybrid approaches have also shown promise. For example, Zhou et al. introduced a two-stream network combining a face classification stream (GoogleNet) with a patch triplet stream for steganalysis-based local noise feature extraction. Another method employed a pairwise-learning model on the CFFN network to capture discriminative differences between GAN-generated and authentic images. [27] [28] These hybrid systems demonstrated improved performance over conventional CNN-only detectors.

1) CNN-based Architecture for Eye and Nose Regions (Model A): Model A is a deep learning network designed with 12 layers, specifically targeting eye and nose features. [29] It is structured using a CNN-based approach, comprising three main blocks, each containing three Conv2D layers with ReLU activation to introduce nonlinearity. To improve training stability and reduce overfitting, the model integrates batch normalization, max pooling, and dropout layers.

Input images are resized to 50×50 pixels. The dataset is divided into 80% for training and 20% for testing. Convolutional layers use a kernel size of (3,3), pooling layers use (2,2), and dropout is set at 0.3. Following the convolutional blocks, a fully connected dense layer with 512 units is included, followed by another dropout layer. [30] [29] The output layer consists of two neurons with a softmax activation for binary classification.

The model is optimized using the Adam optimizer with a learning rate of 0.0001 and trained over 100 epochs. The sparse categorical cross-entropy loss function is used to guide training. [30] A schematic of Model A is shown in Fig. 5, accompanied by its pseudocode.

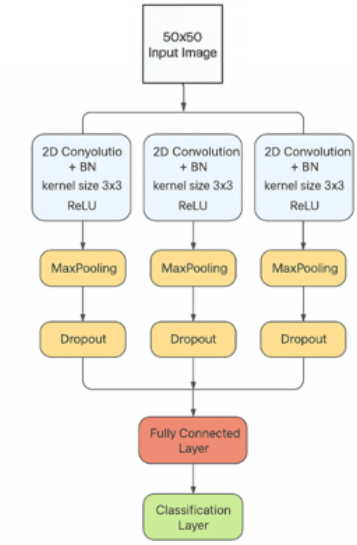

Fig. 3. CNN-based architecture(Model A)

2) CNN-based Architecture for Eye and Nose Regions (Model B): Model B is a lightweight alternative to Model A, built with three Conv2D blocks (six layers total) using ReLU activation, Max Pooling, and Dropout for regularization. [31]Images (50×50) of eye and nose regions are used, with an 80:20 train-test split.

Unlike Model A, batch normalization is omitted, making the model simpler. Training runs for 150 epochs (eye) and 200 epochs (nose). [31] Intermediate feature maps (512 channels) are processed with an eight-head self-attention mechanism and a multilayer perceptron (2048 dimensions). [32] [33] The

Algorithm 1 Model A: CNN with batch normalization and dropout

Input: Image $50 \times 50 \times 3$
Conv Block 1:
Conv layer (3 filters, 3×3, ReLU, pad)
Batch normalization
Conv layer (32 filters, 3×3, ReLU, pad)
Batch normalization
Conv layer (32 filters, 3×3, ReLU, pad)
Batch normalization
Max pooling (2×2, stride 2)
Dropout (0.3)
Conv Block 2: (same as Block 1)
Conv Block 3: (same as Block 1)
Flatten output
Dense (512, ReLU)
Dropout (0.3)
Dense (2, softmax)

architecture has six repeated layers with a weight decay of $1e^{-7}$ to reduce overfitting.

Optimization uses Cross-Entropy loss with batch size 32. The network combines a feature learning module (17 conv layers) and a Vision Transformer (patch-based encoding). [34] Predictions from eye, nose, and face pathways are fused through majority voting to improve deepfake detection accuracy.

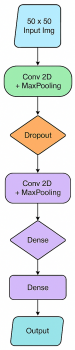

Fig. 4. CNN-based architecture(Model B)

D. Video Detection Models

In recent years, deep learning techniques have achieved remarkable success in detecting forged images. However, extending these methods directly to videos is challenging due to significant loss of frame-level information caused by video compression. [35] [34]To address this, recent research in deepfake video detection can be broadly categorized into biological signal analysis and spatial-temporal feature analysis.

1) Biological Signal Analysis: Researchers have explored natural physiological cues as reliable indicators for deepfake detection. Li utilized eye-blinking patterns, combining CNN and RNN models to capture subtle ocular dynamics, with a

Algorithm 2 Model B: Simple convolutional neural network

Input: Image of size $50 \times 50 \times 3$
Convolutional layer (32 filters, 3×3 kernel, ReLU activation)
Max pooling (2×2 pool size)
Dropout (rate = 0.3)
Convolutional Layer 2:
Replicate structure for Convolutional Layer 2
Convolutional Layer 3:
Replicate structure for Convolutional Layer 3
Flatten the output of the convolutional layers
Dense layer (512 units, ReLU activation)
Dropout (rate = 0.3)
Dense layer (2 units, softmax activation)

binary classifier distinguishing open and closed eyes. This approach, tested on a custom eye-blinking dataset, showed strong effectiveness against fakes. [36]

Similarly, Ciftci et al.leveraged heartbeat signals through a GAN-based framework. Their system extracted facial regions of interest and photoplethysmography (PPG) signals from videos, achieving 97.3% accuracy on public datasets.

Beyond visual cues, audio–visual consistency has also proven valuable. Mittal et al. [41] introduced a Siamese network that jointly analyzes speech and facial movements, using triplet loss on embeddings from OpenFace and pyAudioAnalysis. Their method reached 96.6% accuracy on DeepfakeTIMIT and 84.4% on DFDC, underscoring the strength of multimodal strategies. [38]

2) Spatial and Temporal Feature Analysis: Most conventional deepfake detectors analyze single frames, which can miss subtle manipulations that occur across sequences. Recent approaches focus on temporal consistency between frames to detect video forgery more reliably. A temporally-aware model [8] first extracts frame-level features using a CNN, which are then processed through an LSTM layer to model temporal dependencies in facial expressions and movements. [36]Finally, a softmax layer classifies videos as real or fake. Experiments on a curated set of 600 videos demonstrated the model's efficacy in deepfake detection.

IV. Research Gaps, Challenges, and Future Perspectives

Although CNN- and RNN-based methods, as well as hybrid architectures, have significantly advanced deepfake detection in images and videos, there are several research gaps and challenges that require attention to enhance performance, robustness, and real-world applicability.

A. Research Gaps

Limited Cross-Dataset Generalization: Most CNN- and RNN-based models are trained and validated on specific datasets, reducing their effectiveness on unseen deepfake types or newly emerging manipulations. [37] [35]

Detection of High-Fidelity Deepfakes: High-resolution GAN or RNN-generated deepfakes with subtle facial or temporal artifacts remain difficult to detect due to minimal pixel-level inconsistencies. [38]

Underutilized Multi-Modal Features: While CNNs capture spatial facial features and RNNs model temporal dependencies, integrating audio cues, physiological signals (e.g., eye blinking, heartbeat), or multi-modal embeddings is still underexplored. [38]

Temporal Dynamics in Videos: RNN-based models often focus on short temporal sequences, which limits the detection of long-term manipulations like micro-expressions, lip-syncing, or subtle gesture alterations. [37] [38]

Explainability Deficiency: Many deepfake detection models act as black boxes, lacking interpretability for forensic analysis, which is critical for legal, security, or media verification purposes. [39]

B. Challenges

- Real-Time Video Analysis: CNN-RNN or hybrid models are computationally intensive, making real-time deepfake detection for live streaming challenging. [40]
- Evasion by Adversarial Attacks: Advanced deepfake generators can exploit weaknesses in both CNN and RNN architectures, necessitating robust adversarial defenses. [34]
- Scarcity of Labeled Data for Emerging Manipulations: New deepfake techniques often lack large-scale annotated datasets, limiting the training and validation of effective detection models. [40] [35]
- Cross-Domain and Cross-Platform Limitations: Models trained on a particular dataset (e.g., social media videos) may underperform on different domains or formats, highlighting the need for domain-agnostic detection approaches. [36]

C. Future Directions

- Hybrid Multi-Modal Models: Combining CNNs for spatial feature extraction with RNNs for temporal modeling, alongside audio and physiological signals, can enhance robustness and accuracy. [40]
- Self-Supervised and Few-Shot Learning: Techniques that enable models to learn from limited labeled data or quickly adapt to new deepfake methods can overcome dataset scarcity issues. [34]
- Explainable Deepfake Detection: Developing models that highlight manipulated regions or temporal inconsistencies will improve interpretability for forensic and legal applications.
- Real-Time and Edge Deployment:Optimizing CNN-RNN architectures for lightweight inference on edge devices or streaming platforms is essential for scalable detection. [35]
- Adversarially Robust Models:Integrating adversarial training or defense mechanisms into CNN-RNN hybrids to resist evolving deepfake generation strategies. [36]

V. Results

- CNN (Eye vs. Nose):Eye-based CNN outperformed nose-based CNN, showing that eyes carry stronger cues (blinking, reflections, artifacts) for deepfake detection. [37]
- RNN Temporal Analysis:Outperformed CNNs by capturing frame-to-frame inconsistencies, proving temporal coherence is vital.
- Hybrid CNN+RNN:Achieved the best results, combining CNN's spatial precision with RNN's temporal awareness, leading to superior accuracy, precision, recall, and F1-score. [26]
- Metrics:Accuracy shows overall correctness; precision minimizes false positives; recall captures subtle fakes; F1 ensures balance between the two.
- Confusion Matrices:CNNs showed some misclassifications; RNN reduced errors; hybrid achieved the clearest real–fake separation. [28]

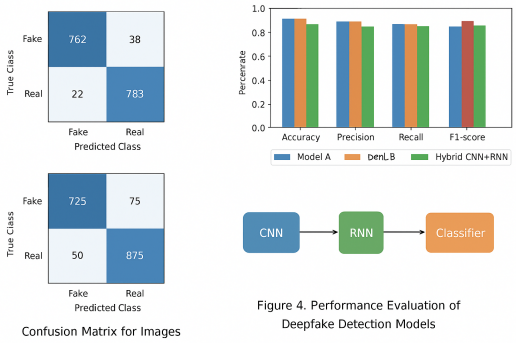

Fig. 5. Result

VI. Conclusion

In this study, we explored the detection of deepfake images and videos using advanced CNN-, RNN-, and hybrid-based architectures. Through a comprehensive analysis of spatial features in images and temporal dynamics in videos, our research demonstrates that integrating convolutional and recurrent networks can significantly enhance the detection accuracy of manipulated media. [28] The comparative evaluation of different models for eye, nose, and facial regions, as well as the consideration of physiological and multi-modal signals, emphasizes the potential of hybrid approaches in capturing subtle inconsistencies present in deepfakes.

Despite the remarkable progress in deepfake detection, challenges such as cross-dataset generalization, high-fidelity manipulations, real-time processing, and explainability remain prevalent. [35] Addressing these challenges through multi-modal, adaptive, and interpretable frameworks is crucial to stay ahead of rapidly evolving deepfake generation techniques.

Overall, our work highlights the importance of combining CNN-based spatial analysis with RNN-driven temporal modeling, demonstrating a robust pathway for reliable deepfake detection. The proposed frameworks not only achieve high

accuracy on benchmark datasets but also provide a foundation for developing scalable, real-world detection systems that can safeguard digital media integrity in an era of increasingly sophisticated synthetic content. [40]

REFERENCES

[1] Nataraj, L., et al. (2019). Detecting GAN Generated Fake Images Using Co-Occurrence Matrices. *Electronic Imaging*, 2019, 532-1–532-7. https://doi.org/10.2352/ISSN.2470-1173.2019.5.MWSF-532

[2] Wang, S.-Y., Wang, O., Zhang, R., Owens, A. and Efros, A.A. (2020). CNN-Generated Images Are Surprisingly Easy to Spot... for Now. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Seattle, 13–19 June 2020, pp. 8695–8704. https://doi.org/10.1109/CVPR42600.2020.00872

[3] Mitra A, Mohanty SP, Corcoran P, Kougianos E (2021) A machine learning based approach for deepfake detection in social media through key video frame extraction. SN Comput Sci 2:1–18

[4] Mogan, J.N., Lee, C.P., Lim, K.M., Ali, M., and Alqahtani, A. (2023). GaitCNN-ViT: Multi-model gait recognition with convolutional neural networks and vision transformer. *Sensors*, 23(8), 3809.

[5] Yang, Z., Liang, J., Xu, Y., Zhang, X.Y., and He, R. (2023). Masked relation learning for deepfake detection. *IEEE Transactions on Information Forensics and Security*, 18, 1696–1708.

[6] Al-Dhabi, Y., and Zhang, S. (2021). Deepfake video detection by combining convolutional neural network (CNN) and recurrent neural network (RNN). In *Proceedings of the 2021 IEEE International Conference on Computer Science, Artificial Intelligence and Electronic Engineering (CSAIEE)*.

[7] Anon. (2019). Detecting and simulating artifacts in GAN-fake images. In *Proceedings of the IEEE International Workshop on Information Forensics and Security (WIFS)*, Dec. 2019, pp. 1–6. Retrieved June 18, 2023, from Google Search link.

[8] Patel, Y., Tanwar, S., Bhattacharya, P., Gupta, R., Alsuwian, T.M., Davison, I.E., and Mazibuko, T.Z.F. (2023). An improved dense CNN architecture for deepfake image detection. *IEEE Access*. https://doi.org/10.1109/ACCESS.2023.3251417

[9] Fakeapp. (2018). *Fakeapp software*. Available at: https://www.fakeapp.org/ (Accessed: May 29, 2018).

[10] IEEE's Signal Processing Society. (2018). Camera Model Identification — Kaggle. Available at: https://www.kaggle.com/c/sp-society-camera-model-identification/discussion/49299 (Accessed: May 29, 2018).

[11] Rana, M.S., and Sung, A.H. (2020). DeepfakeStack: A Deep Ensemble-based Learning Technique for Deepfake Detection. *Proceedings of CSCloud-EdgeCom 2020*. doi:10.1109/CSCloud-EdgeCom49738.2020.00021.

[12] El Rai, M.C., Al Ahmad, H., Gouda, O., Jamal, D., Talib, M.A., and Nasir, Q. (2020). Fighting Deepfake by Residual Noise Using Convolutional Neural Networks. *Proceedings of ICSPIS 2020*. doi:10.1109/ICSPIS51252.2020.9340138.

[13] Kolagati, S., Priyadharshini, T., and Mary Anita Rajam, V. (2022). Exposing deepfakes using a deep multilayer perceptron – convolutional neural network model. *International Journal of Information Management Data Insights*, 2(1), 100054. doi:10.1016/j.jjimei.2021.100054.

[14] Botelho De Souza, G., Da Silva Santos, D.F., Goncalves Pires, R., Papa, J.P., and Marana, A.N. (2018). Efficient width-extended convolutional neural network for robust face spoofing detection. *Proceedings of BRACIS 2018*. doi:10.1109/BRACIS.2018.00047.

[15] Wodajo, E.; Hossain, M.S.; Kim, J. Convolutional vision transformer for deepfake detection. IEEE Access 2021, 9, 125634–125643.

[16] Tran, D.T.; Tran, T.T.; Le, T.L. A deepfake detection model for altered video with content-based approach. Int. J. Adv. Comput. Sci. Appl. 2021, 12, 104–112.

[17] Ismail, M.; Mahmood,M.T.;Wahab,A.W.Deepfake detection using You Only Look Once Convolution Recurrent Neural Networks. Neural Comput. Appl. 2021, 33, 13643–13656.

[18] Mitra, S.; Roy, A.; Dubey, S. Deepfake video detection through visual artifact analysis. Multimed. Tools Appl. 2021, 80, 25917–25933.

[19] Ge,Y.; Huang, J.; Zhu, X.; Long, M. Self-distillation fine-tuning for DeepFake video detection. IEEE Trans. Inf. Forensics Secur. 2021, 16, 3045–3058.

[20] Liu, Y.; Wang, J.; He, Q. A portable 3D convolutional neural network for deepfake detection. Neurocomputing 2021, 457, 176–188.

[21] Heo,J.; Kang, M.J.; Lee, S. Vision transformer for deepfake detection with content-based patch embedding. IEEE Trans. Circuits Syst. Video Technol. 2021, 32, 950–964.

[22] Kaliyar, R.K., Goswami, A., and Narang, P. (2021). DeepFakE: Improving fake news detection using tensor decomposition-based deep neural network. *Journal of Supercomputing*. doi:10.1007/s11227-020-03294-y.

[23] Nichol, A., Dhariwal, P., Ramesh, A., Shyam, P., Mishkin, P., McGrew, B., Sutskever, I., and Chen, M. (2022). GLIDE: Towards photorealistic image generation and editing with text-guided diffusion models. *arXiv preprint arXiv:2112.10741*.

[24] Yi, J., Wang, C., Tao, J., Zhang, X., Zhang, C.Y., and Zhao, Y. (2023). Audio deepfake detection: A survey. *arXiv preprint arXiv:2308.15252*.

[25] Tan, C., Tao, R., Liu, H., and Zhao, Y. (2024). GANGen-Detection: A dataset generated by GANs for generalizable deepfake detection. *Dataset Release*. Available at: https://github.com/chuangchuangtan/GANGen-Detection.

[26] Mittal, G., Hegde, C., and Memon, N. (2023). Gotcha: Real-time video deepfake detection via challenge-response. *arXiv preprint arXiv:2304.03787*.

[27] Yan, Z., Yao, T., Chen, S., Zhao, Y., Fu, X., Zhu, J., Luo, D., Yuan, L., Wang, C., Ding, S., et al. (2024). DF40: Toward next-generation deepfake detection. *arXiv preprint arXiv:2406.13495*.

[28] Snopes Media Group Inc. (2025). About Snopes. Snopes is a widely trusted fact-checking website that employs professional journalists who investigate viral claims and urban legends. Available at: https://www.snopes.com.

[29] Agence France-Presse. (2025). About AFP Fact Check. AFP Fact Check is the fact-checking division of Agence France-Presse, a major world news agency. Available at: https://factcheck.afp.com.

[30] iProov. (2024). Threat Intelligence Report 2024: The Impact of Generative AI on Remote Identity Verification. *Technical Report*. iProov. Available at: https://www.iproov.com/resources/threat-intelligence-report-2024.

[31] A. Heidari, N. Jafari Navimipour, H. Dag, and M. Unal, "Deepfake detection using deep learning methods: A systematic and comprehensive review," *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, vol. 14, p. e1520, 2024.

[32] R. U. Maheshwari, S. Kumarganesh, S. KVM, A. Gopalakrishnan, K. Selvi, B. Paulchamy, P. Rishabavarthani, K. M. Sagayam, B. K. Pandey, and D. Pandey, "Advanced plasmonic resonance enhanced biosensor for comprehensive real-time detection and analysis of deepfake content," *Plasmonics*, pp. 1–18, 2024.

[33] A. Heidari, N. J. Navimipour, H. Dag, S. Talebi, and M. Unal, "A novel blockchain-based deepfake detection method using federated and deep learning models," *Cognitive Computation*, pp. 1–19, 2024.

[34] P. Singh and D. B. Dhiman, "Exploding AI-generated deepfakes and misinformation: A threat to global concern in the 21st century," *Available at SSRN 4651093*, 2023.

[35] W. Matli, "Extending the theory of information poverty to deepfake technology," *International Journal of Information Management Data Insights*, vol. 4, p. 100286, 2024.

[36] A. O. Kwok and S. G. Koh, "Deepfake: a social construction of technology perspective," *Current Issues in Tourism*, vol. 24, pp. 1798–1802, 2021.

[37] D. Chapagain, N. Kshetri, and B. Aryal, "Deepfake disasters: A comprehensive review of technology, ethical concerns, countermeasures, and societal implications," in *Proc. 2024 International Conference on Emerging Trends in Networks and Computer Communications (ETNCC)*, IEEE, pp. 1–9, 2024.

[38] D. Sarkar and S. De Sarkar, "Combatting deep-fakes in India – an analysis of the evolving legal paradigm and its challenges," 2024.

[39] M. Mustak, J. Salminen, M. Mäntymäki, A. Rahman, and Y. K. Dwivedi, "Deepfakes: Deceptions, mitigations, and opportunities," *Journal of Business Research*, vol. 154, p. 113368, 2023.

[40] M. R. Shoaib, Z. Wang, M. T. Ahvanooey, and J. Zhao, "Deepfakes, misinformation, and disinformation in the era of frontier AI, generative AI, and large AI models," in *Proc. 2023 International Conference on Computer and Applications (ICCA)*, IEEE, pp. 1–7, 2023.